

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Systems for Data Science, Optimization in CS, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, SQL, JavaScript, TypeScript, C++, Java, Bash

ML & Data Science

pandas, NumPy, scikit-learn, predictive modeling, statistical analysis, transfer learning, PyTorch, JAX, HuggingFace Transformers, sentence-transformers

NLP & LLM Systems

RAG, LLM agents, RRF, cross-encoder reranking, MMR, knapsack token budgeting, OpenAI / Groq APIs, Ollama, llama-index, spaCy

Backend & Data Eng

Apache Spark / PySpark, Databricks, FastAPI, Node.js, REST APIs, PostgreSQL, MongoDB, Redis, SQLite

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected a full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data and refine production NLP/clinical models; replaced spreadsheet workflows with secure role-based APIs and reviewer dashboards that cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) processing **10,000+ monthly submissions** for **8+ UK clinical clients**, designing aggregation pipelines and integration paths to Mayden's iaptus platform; contributed to **\$340K+** in revenue.
- Acted as primary technical point of contact for the eTriage backend, collaborating across clinical, AI, and product stakeholders to ship authentication, rate limiting, and audit logging features, and ran knowledge transfer sessions for new developers joining the UK clinical team.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by **PostgreSQL**, and integrated **LLM-powered context-aware replies** into a chat product, iterating on prompt design and response-quality heuristics for a real customer-facing experience.

Projects

AgenticSearch - LLM Pipeline for Insights from Unstructured Web Data

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage **LLM-powered pipeline** that extracts structured insights from unstructured web sources via typed retrieval planning, page rerank, deterministic parsers with LLM fallback, output verification, and **cell-level provenance** for every extracted result.
- Hardened the service with **150+ automated tests**, provider routing/fallback across OpenAI/Groq backends, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

KG2RAG-Enhanced - Multi-Hop QA on HotpotQA

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama, sentence-transformers, llama-index, spaCy, PyTorch | UMass CS 685 Advanced NLP

- Extended KG2RAG with **multi-view seed retrieval** on HotpotQA: generated sub-questions, retrieved passages per view, and fused results via Reciprocal Rank Fusion (RRF), cross-encoder reranking, and Maximal Marginal Relevance (MMR).
- Replaced heuristic top-M selection with a **0-1 knapsack token-budget** formulation (value = relevance × coverage); integrated it into an Ollama + LLaMA-3 pipeline and ran **100-500-question batch experiments** with token and latency logging.

Spark ETL Performance Optimization - NYC TLC Taxi Dataset

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, Databricks, SQL, Python

- Optimized a distributed **Spark/PySpark ETL pipeline** on Databricks (NYC TLC taxi dataset) by tuning broadcast joins, caching, partitioning, adaptive Spark settings, and column pruning to reduce shuffle volume and wall-clock runtime across millions of trip records.
- Profiled stage-level timings via **Spark UI** to localize shuffle and skew bottlenecks across naive vs. tuned runs and documented a repeatable optimization workflow for downstream large-scale analytics jobs.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, built and benchmarked deep transfer-learning predictive models on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.